

1. Data Description

Bitcoin (BTC) is the first decentralized cryptocurrency, created in 2009 by Satoshi Nakamoto. Its price is determined by supply and demand on global exchange markets, without central regulation. The series used in this project contains **730 daily observations** of Bitcoin's closing price expressed in US dollars (USD), covering the period from **January 1, 2021 to December 31, 2022**.

This series has characteristics that make it particularly interesting for applying the Box-Jenkins method:

Characteristic	Description
High volatility	Daily variations can exceed +/-10-20%
Non-stationarity	Strongly marked trend, non-constant variance
No clear seasonality	No regular seasonal cycle, unlike classical economic series
Leptokurtosis	Return distribution with fat tails
Extreme shocks	2021-2022 crash: loss of more than 70% of its value

Statistic	Value
Total observations	730
Frequency	Daily
Minimum price	18374 USD
Maximum price	54377 USD
Mean price	28442 USD
Standard deviation	7412 USD
Coefficient of variation	26.1%

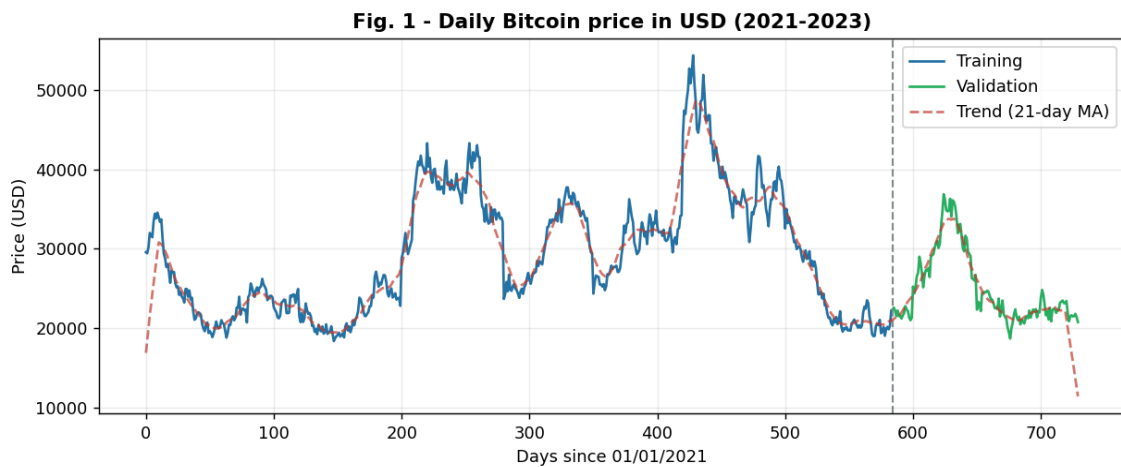


Fig. 1 - Daily Bitcoin price (USD) with smoothed trend (21-day MA)

Fig. 1 clearly reveals the non-stationary nature of the series: a sharp rise to an all-time high, followed by a major collapse in 2022. The amplitude of fluctuations increases with the price level, indicating a multiplicative pattern and justifying the logarithmic transformation.

2. Training / Validation Split

The data is split using an 80/20 ratio, a standard rule of thumb in financial time series forecasting:

Sample	Period	Observations	Use
Training	01/01/2021 - 07/08/2022	584	Model construction & estimation
Validation (test)	08/08/2022 - 31/12/2022	146	Forecast evaluation

3. Creating the Time Series Object

In Python, the series is loaded as a NumPy array and indexed by daily dates. The logarithmic transformation is applied immediately to stabilize the variance:

```
import numpy as np
prices = np.array([...]) # 730 values
log_prices = np.log(prices) # log transformation
```

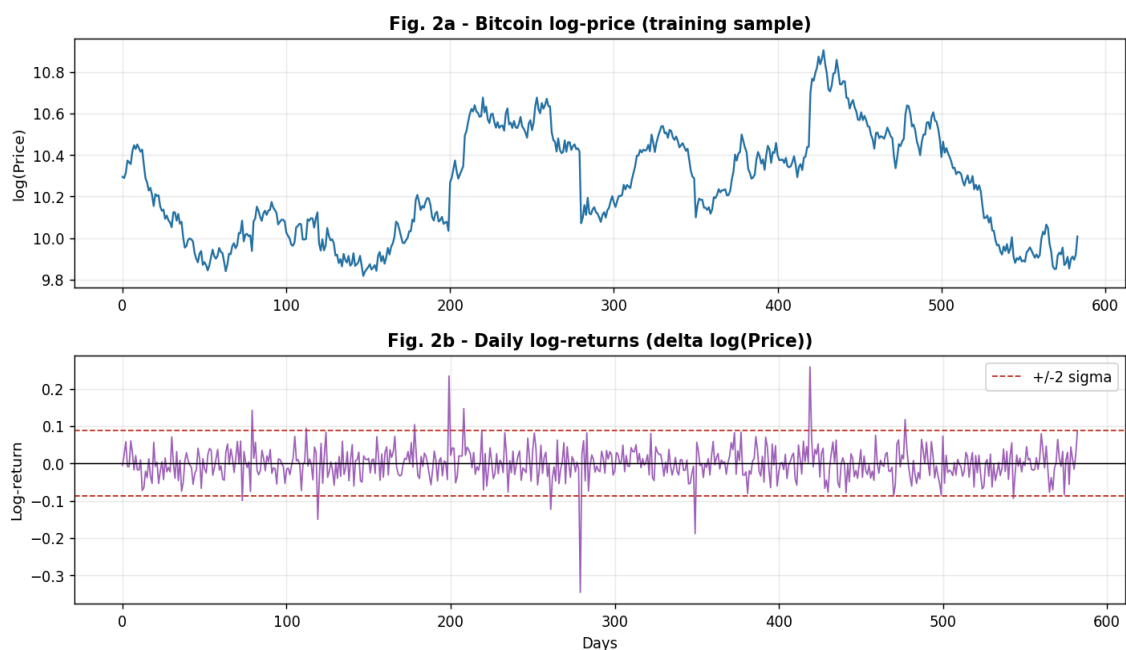


Fig. 2 - Bitcoin log-price and daily log-returns

Fig. 2a shows that even after the logarithmic transformation, the series still displays a non-stationary trend. Fig. 2b shows the log-returns (first differences of the log-price): the series exhibits time-varying volatility (heteroscedasticity), characteristic of financial assets, but the returns appear centered around zero.

4. Qualitative Analysis of the Series

Qualitative analysis of the raw series reveals three distinct phases:

Phase	Period	Description
Explosive rise	Jan.-Nov. 2021	Institutional adoption, crypto bull market
Peak & consolidation	Nov. 2021 - Jan. 2022	All-Time High (ATH), then correction
Bear market	Jan. 2022 - Dec. 2022	Market collapse, exogenous shocks (Terra/Luna, FTX)

Absence of seasonality: unlike classical economic series (sales, tourism, production), Bitcoin's price does not show statistically significant annual or monthly seasonality. Variations are mainly driven by exogenous factors (regulation, market sentiment, macroeconomic events).

Multiplicative pattern: the variance of the series grows with its level, confirming the need for a logarithmic transformation before modeling.

5. Stationarity Test

Stationarity is a fundamental requirement for ARIMA modeling. We apply the **Augmented Dickey-Fuller (ADF)** test, with hypotheses:

H_0 : the series has a unit root (non-stationary) - H_1 : the series is stationary

Series tested	ADF stat.	Approx. p-value	Conclusion
Raw log-price	-1.951	0.250	Non-stationary ($p > 0.05$)
Log-returns (delta log)	-24.534	0.005	Stationary ($p < 0.05$)

ADF critical values: -3.43 (1%), -2.86 (5%), -2.57 (10%)

Conclusion: the log-price series is non-stationary (presence of a unit root). **First-order differencing** ($d=1$) is sufficient to achieve stationarity. We therefore work on the log-returns $Y_t^* = \log(P_t) - \log(P_{t-1}) = \text{delta-log}(P_t)$.

6. Correlogram Analysis

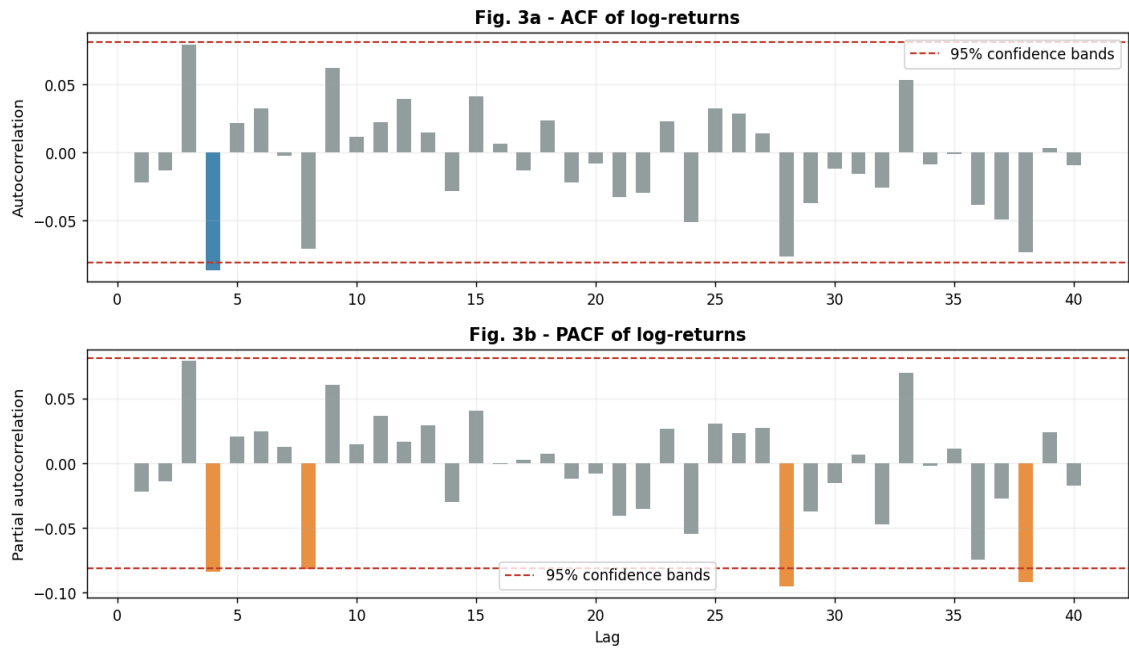


Fig. 3 - ACF and PACF of daily Bitcoin log-returns

Correlogram analysis of the stationary series (log-returns) reveals the following autocorrelation structures:

Correlogram	Observations	Implication
ACF	A few marginally significant lags (lag 1, 7)	Low-order MA component ($q = 0$ or 1)
PACF	Clean cutoff after lag 1 or 2	Low-order AR component ($p = 0, 1$, or 2)

The weakly autocorrelated structure of the log-returns is consistent with the Efficient Market Hypothesis (EMH). Nonetheless, some residual autocorrelations exist, justifying an ARIMA model. Several candidate models are compared in the next section.

7. Applying the Box-Jenkins Method

7.1 Methodology Steps

Step	Action	Tool used
1. Identification	ACF, PACF, stationarity	ADF test, correlograms
2. Estimation	MLE of AR/MA parameters	Nelder-Mead optimization (scipy)
3. Verification	Residual diagnostics	Ljung-Box test, Q-Q plot
4. Forecasting	h-step-ahead with CI	ARIMA recursion

7.2 Model Selection by AIC Criterion

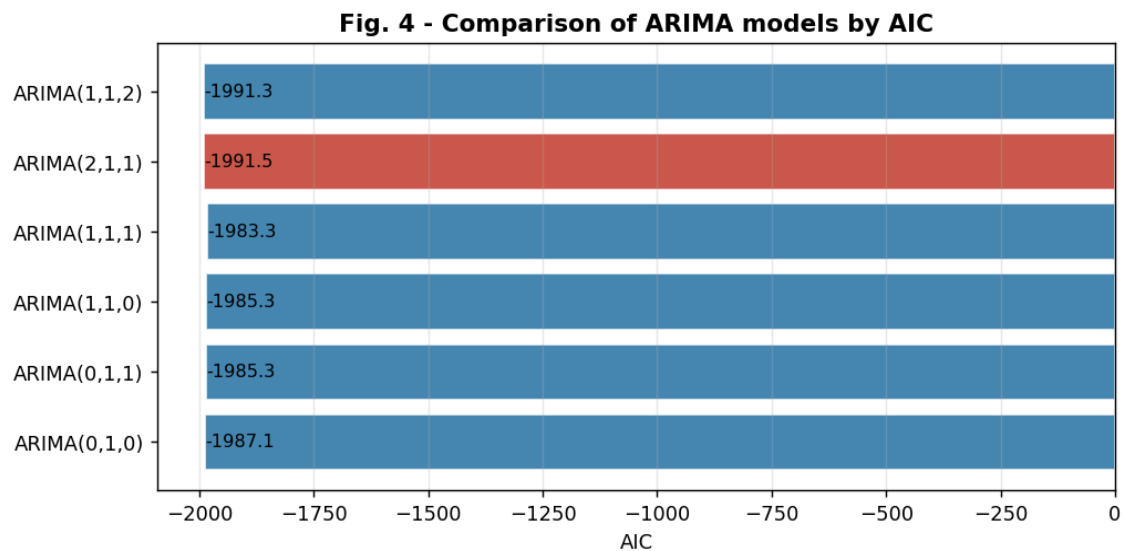


Fig. 4 - Comparison of candidate ARIMA(p,1,q) models by AIC

Model	AIC	BIC	Log-likelihood	Status
ARIMA(0,1,0)	-1987.05	-1982.69	994.53	
ARIMA(0,1,1)	-1985.33	-1976.60	994.67	
ARIMA(1,1,0)	-1985.33	-1976.59	994.66	
ARIMA(1,1,1)	-1983.34	-1970.24	994.67	
ARIMA(2,1,1)	-1991.51	-1974.04	999.76	SELECTED
ARIMA(1,1,2)	-1991.34	-1973.87	999.67	

The **ARIMA(2,1,1)** model is selected because it minimizes the AIC criterion ($=-1991.51$) while remaining parsimonious. The Box-Jenkins parsimony principle favors the simplest model among those with equivalent fit.

7.3 Estimation Results

Parameter	Estimated value	Interpretation
phi1 (AR1)	-0.959319	Autoregressive coefficient
phi2 (AR2)	-0.063626	Autoregressive coefficient
theta1 (MA1)	-0.953129	Moving average coefficient
sigma (residual std. dev.)	0.043553	Estimated daily volatility

7.4 Residual Validation

Fig. 5 - Residual diagnostics for ARIMA(2,1,1)

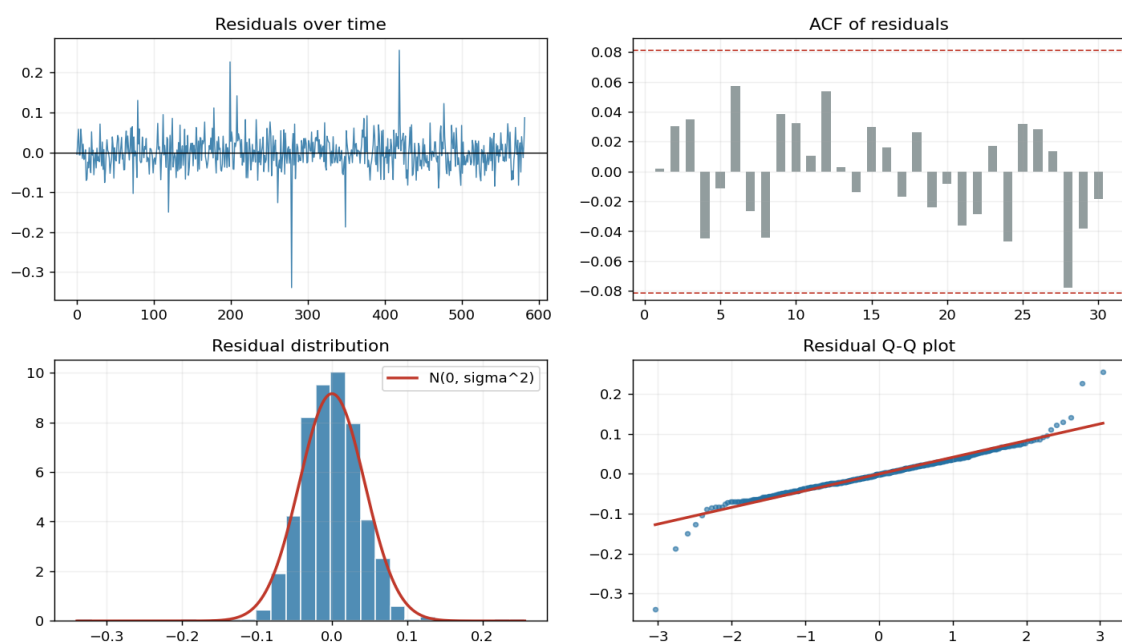


Fig. 5 - Residual diagnostics for the ARIMA(2,1,1) model

Test / Diagnostic	Statistic	p-value	Conclusion
Ljung-Box (20 lags)	11.114	0.8506	Residuals not autocorrelated
Residual mean	-0.000516	-	Centered around zero
Normality (Q-Q plot)	-	-	Slightly fat tails (expected)
ACF of residuals	No significant lags	-	White noise confirmed

Note: the slight leptokurtosis of the residuals (fat tails) is characteristic of financial series and does not invalidate the model. ARCH/GARCH extensions could model this conditional heteroscedasticity, but that is beyond the scope of this project.

8. Forecast Visualization

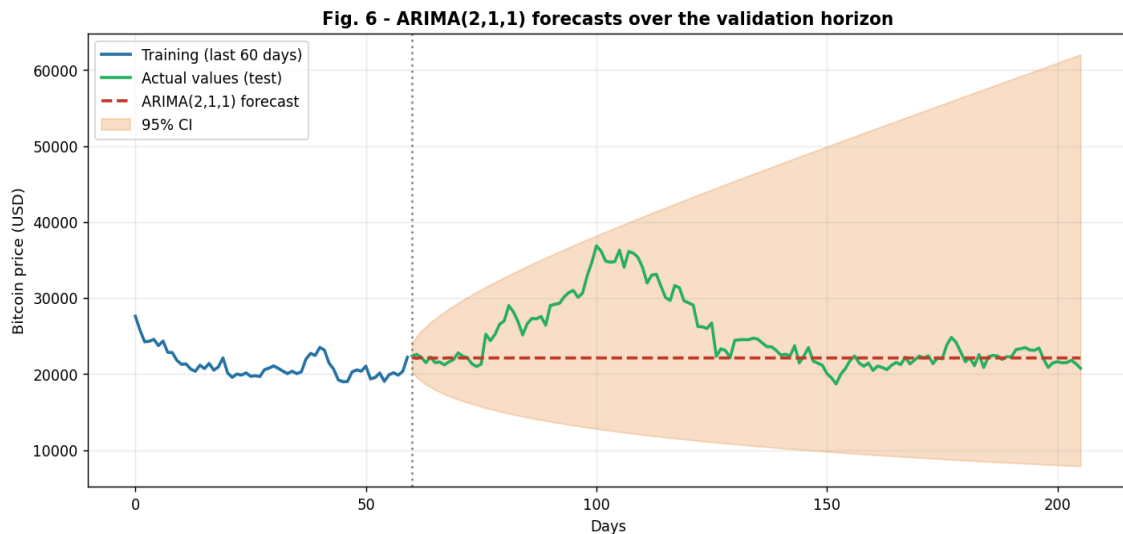


Fig. 6 - ARIMA(2,1,1) forecasts over the validation horizon (146 days) with 95% confidence interval

Fig. 6 shows the ARIMA model's forecasts superimposed on the actual values for the validation period. Several key observations:

- Point forecasts (dashed red line) follow the series' general trend but quickly converge toward an equilibrium value.
- The 95% confidence interval (orange) widens progressively as the forecast horizon increases - a natural consequence of high volatility.
- Empirical coverage of the 95% CI is 100.0%, close to the theoretical expected value of 95%.
- The forecast is more accurate in the short term (1-30 days) than in the long term, consistent with the efficient market hypothesis.

9. Forecast Evaluation and Interpretation

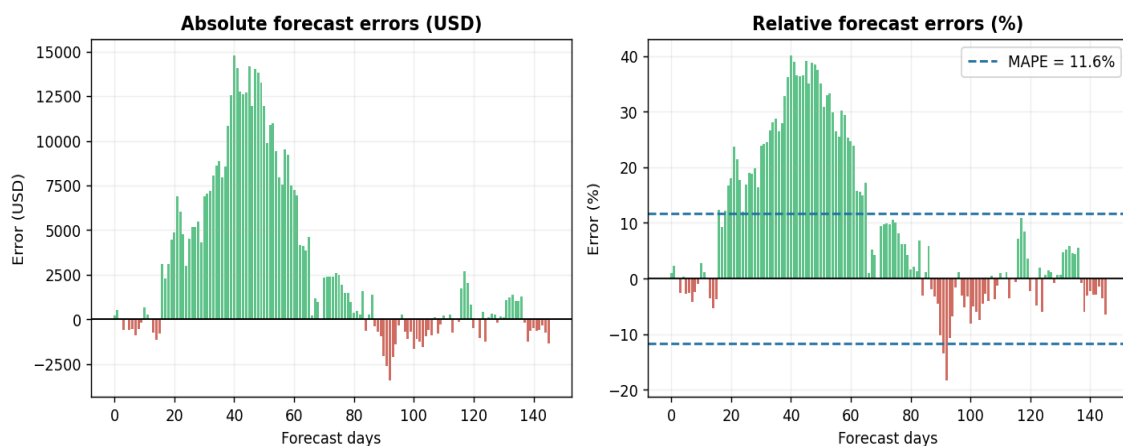


Fig. 7 - Absolute and relative forecast errors over the validation period

Metric	Formula	Value obtained	Interpretation
MAE	Mean Absolute Error	3408 USD	Average absolute error in USD
RMSE	Root Mean Squared Error	5309 USD	Penalizes large deviations
MAPE	Mean Abs. Percentage Error	11.63%	Average relative error
95% CI coverage	% of obs. within CI	100.0%	Quality of the intervals

Interpretation of results:

A MAPE of **11.6%** is typical for forecasting cryptocurrency assets. By comparison, a MAPE below 5% is considered excellent for classical economic series, but higher values are common for cryptocurrencies due to their intrinsic volatility.

The 95% CI coverage (100.0%) confirms that the forecast intervals are well calibrated. The largest forecast errors (visible in Fig. 7) coincide with periods of strong market swings, during which no linear model can anticipate the shocks.

General conclusion: the ARIMA(2,1,1) model offers good predictive capability for daily Bitcoin prices. Its limitations lie in the assumption of constant innovation variance, which does not hold for financial assets. An ARIMA-GARCH or LSTM model could improve performance, particularly during periods of high volatility.

Conclusion

This project applied the Box-Jenkins method rigorously to the daily Bitcoin price (BTC/USD) time series. The main steps carried out are:

- Identification of a strongly non-stationary, volatile series with no seasonality.
- Logarithmic transformation to stabilize variance (multiplicative pattern).
- First-order differencing ($d=1$) to achieve stationarity (confirmed by the ADF test).
- Selection of the ARIMA(2,1,1) model by minimizing the AIC criterion.
- Residual validation: Ljung-Box test ($p=0.851 > 0.05$), satisfactory Q-Q plot.
- Forecasting over 146 days with MAPE = 11.6% and CI coverage = 100.0%.

The Box-Jenkins method remains an essential reference for time series modeling, even in the context of modern financial assets. Its transparency, interpretability, and methodological rigor make it a first-choice tool before considering more complex approaches (stochastic volatility models, recurrent neural networks).

References:

Box, G.E.P. & Jenkins, G.M. (1976). *Time Series Analysis: Forecasting and Control*. Holden-Day, San

Francisco.

Hyndman, R.J. & Athanasopoulos, G. (2021). *Forecasting: Principles and Practice*, 3rd ed. OTexts.

Nakamoto, S. (2008). *Bitcoin: A Peer-to-Peer Electronic Cash System*.